

Technical Document #198: Cloud Computing - Comprehensive Overview

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Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet.
- Multi-cloud strategies use services from multiple cloud providers.
- Content delivery networks (CDNs) distribute content globally to reduce latency.